

SONNBLICK, Austria

WMO: 113430

Lat: 47.054N

Lon: 12.957E

Elev: 3114

StdP: 69.10

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.3	-21.4	-29.9	0.3	-14.3	-27.2	0.5	-12.1	23.2	-9.7	21.6	-10.2	9.7	30	1.155

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	10.2	6.7	9.0	6.3	8.0	5.7	7.9	9.4	7.0	8.4	6.2	7.6	4.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	9.3	8.6	6.2	8.7	7.7	5.4	8.2	6.9	32.4	9.4	30.1	8.5	28.0	7.7	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.4	16.3	DB	-25.9	13.3	2.9	3.3	-27.9	15.7	-29.6	17.6	-31.3	19.5	-33.4	21.9	(4)
(5)			WB	-26.0	11.1	3.0	3.4	-28.2	13.5	-29.9	15.6	-31.6	17.5	-33.7	20.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.6	-11.6	-12.2	-10.1	-6.6	-2.8	1.4	3.4	3.3	-0.5	-2.7	-6.9	-10.0	(6)	
	DBStd	6.82	4.47	4.81	4.60	3.90	3.37	3.79	3.30	3.18	3.50	3.88	4.66	4.77	(7)	
	HDD10.0	5316	670	621	623	499	396	259	205	207	314	395	506	621	(8)	
	HDD18.3	8357	928	854	882	749	654	509	463	465	564	653	756	879	(9)	
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	7.8	9.3	9.5	8.5	7.8	7.3	6.1	6.1	5.9	6.6	8.6	9.6	9.0	(14)	
(15) Precipitation	PrecAvg	1711	136	116	153	160	156	150	164	166	123	123	141	140	(15)	
	PrecMax	2100	320	264	344	313	492	225	257	357	261	255	321	281	(16)	
	PrecMin	1250	27	28	40	22	33	71	55	70	53	1	9	12	(17)	
	PrecStd	229	67	58	65	63	79	37	45	51	44	62	65	58	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.3	-0.9	-0.4	2.1	6.8	11.5	11.9	11.5	7.8	5.7	2.7	0.3	(19)	
		MCWB	-5.1	-6.2	-4.6	-0.5	3.4	6.9	8.0	7.4	4.6	0.5	-2.1	-4.5	(20)	
	2%	DB	-2.8	-2.5	-2.0	0.4	4.8	9.4	10.4	9.8	6.4	4.4	1.2	-1.5	(21)	
		MCWB	-6.2	-6.7	-5.0	-1.9	2.3	5.9	7.4	6.2	3.6	-0.2	-2.4	-6.0	(22)	
	5%	DB	-4.0	-4.4	-3.2	-0.8	3.3	8.0	9.1	8.7	5.3	3.3	0.1	-2.9	(23)	
		MCWB	-6.6	-7.0	-5.9	-2.4	1.3	5.3	6.6	6.0	2.9	-0.8	-3.8	-6.6	(24)	
	10%	DB	-5.6	-6.1	-4.3	-1.9	1.8	6.6	8.0	7.7	4.2	2.2	-1.4	-4.1	(25)	
		MCWB	-7.7	-7.8	-6.3	-3.1	0.7	4.5	5.9	5.6	2.2	-1.1	-4.1	-6.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.6	-4.3	-2.8	0.1	4.5	8.3	9.5	8.7	6.1	2.8	0.5	-3.1	(27)	
		MCDB	-2.7	-3.4	-1.9	1.1	6.0	10.3	11.0	9.8	7.1	3.8	1.2	-1.4	(28)	
	2%	WB	-5.1	-5.6	-4.0	-0.9	3.0	6.9	8.1	7.6	4.8	1.7	-1.3	-4.4	(29)	
		MCDB	-3.7	-4.0	-3.0	0.0	4.3	8.5	9.7	8.9	5.7	2.7	-0.2	-3.4	(30)	
	5%	WB	-6.3	-6.7	-5.0	-1.9	1.8	5.9	7.2	6.8	3.8	0.9	-2.3	-5.2	(31)	
		MCDB	-4.7	-4.9	-3.9	-1.1	2.9	7.5	8.7	8.1	4.7	1.9	-1.3	-4.0	(32)	
	10%	WB	-7.3	-7.7	-5.9	-2.8	0.8	4.9	6.3	6.0	2.9	0.0	-3.1	-6.2	(33)	
		MCDB	-5.8	-6.0	-4.7	-2.0	1.6	6.5	7.7	7.3	3.8	1.1	-2.1	-4.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.1	4.2	4.1	3.9	4.1	4.3	3.9	3.4	3.5	3.6	4.1	(35)	
		MCDBR	5.0	4.7	4.1	3.9	4.7	5.5	5.4	4.6	4.0	3.5	3.8	4.6	(36)	
	5% WB	MCWBR	3.6	3.7	3.6	3.8	4.3	5.1	4.9	4.4	4.6	3.9	2.6	3.4	(37)	
		MCWBR	4.5	5.1	3.9	3.8	4.5	5.2	5.2	4.4	3.8	3.1	3.0	4.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.193	0.208	0.236	0.263	0.274	0.282	0.281	0.277	0.257	0.240	0.218	0.195	(40)	
		taud	2.480	2.443	2.468	2.517	2.543	2.546	2.542	2.570	2.624	2.648	2.628	2.546	(41)	
	Ebn at Noon		963	1003	1010	1001	992	981	978	973	974	953	925	927	(42)	
		Edh at Noon	60	78	91	97	98	99	98	91	78	64	53	50	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.38	2.27	3.31	4.16	4.80	5.35	5.45	4.81	3.56	2.43	1.39	1.14	(44)	
	RadStd		0.16	0.29	0.40	0.54	0.58	0.50	0.47	0.53	0.45	0.27	0.24	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (88 neighbors)	+0.43	N/A	N/A	+0.63	+0.39	+0.33	N/A	N/A	+72	+40

Nomenclature: See separate page